Product Requirement Document (PRD): Enterprise Sovereign LLM Server

ON-PREMISE 7B MULTILINGUAL LLM INFRASTRUCTURE SPECIFICATION

1. Project Executive Overview

This document defines the deployment specification for an on-premise conversational AI server utilizing a 7 Billion parameter multilingual model with strict data residency requirements.

2. Technical & Hardware Constraints Matrix

Parameter	Specification Value	Architectural Note
Model Architecture	Indic-Mistral-7B Large Language Model	Transformer decoder with grouped-query attention
Model Parameter Count	7 Billion parameters (7.0B)	7B dense parameters for multilingual Hindi/Tamil/Telugu processing
Weight Precision	FP16 (Half Precision)	16-bit floating point for optimal generation perplexity
Target Output Time / TTFT	45 ms (Time to First Token)	Sub-50ms conversational responsiveness for enterprise chatbot
Target Throughput	60 tokens per second (tok/s)	Real-time streaming text generation for user interactions
Target Budget	INR 2,50,000 (Rs. 250,000)	Workstation / on-premise edge server investment
Thermal Power Envelope	450 Watts TDP	Standard dual-slot PCIe workstation cooling infrastructure
Context Window Length	4096 tokens	Accommodates long-context document Q&A; and RAG grounding
Concurrent User Streams	8 active streams	Dynamic continuous batching via vLLM inference engine
Deployment Technology	vLLM / Docker Container	OpenAI-compatible HTTP REST endpoint on internal network

3. Deployment & Operational Requirements

This project deploys a sovereign on-premise Large Language Model server for an Indian financial institution. Due to strict data residency and privacy compliance, all customer document summaries must execute on-premise without cloud transmission. The model has 7 Billion parameters running in FP16 precision with a 4096-token context window. Target performance requires streaming throughput of at least 60 tokens per second and Time to First Token (TTFT) under 45 ms. The hardware budget is INR 250,000 (Rs. 2,50,000) with an allowable thermal envelope up to 450 Watts TDP.